

Ultrafast Singlet Fission and Intersystem Crossing in Halogenated Tetraazaperopyrenes

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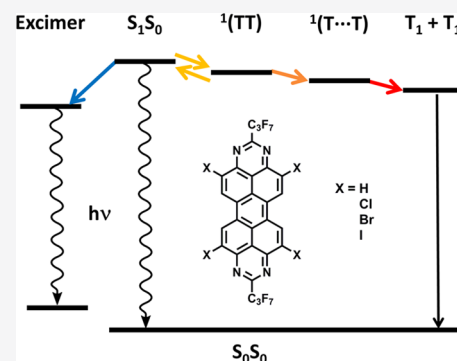
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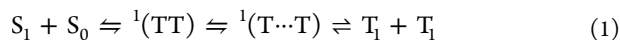
Supporting Information

ABSTRACT: Charge carrier multiplication via singlet fission into two triplet states has the potential to increase efficiencies of photovoltaics by one-third due to the reduction of thermalization losses. In the present work, we investigate tetraazaperopyrenes, a class of *N*-heteropolycycles, as suitable singlet fission candidates. Using a combined experimental and theoretical approach, fundamentally different mechanisms for triplet formation in solution and thin film are identified. In solution, an ultrafast intersystem crossing process is observed, which is accelerated for heavier halide substituents not only due to enhanced spin–orbit coupling but also due to the energy tuning between the S_1 and T_2 states. In thin films, a correlated triplet pair is formed coherently upon photoexcitation. Subsequently, an excimer formation is observed, which competes with the electronic decorrelation of the triplet pair. The comparison with peropyrene shows that aza-substitutions within the aromatic core can be a powerful strategy for tuning the energy levels of the states important to singlet fission.



1. INTRODUCTION

The increasing role of photovoltaics as a sustainable energy source has been driving the investigation of novel approaches to increase conversion efficiencies. One of these approaches is charge carrier multiplication, which has the potential to mitigate thermalization losses present in photovoltaics.^{1–4} This has the possibility to surpass the Shockley–Queisser limit, raising the theoretical efficiency limit from 33 to 44%.^{5–9} Among all multiplication mechanisms, singlet fission (SF) is a promising spin-allowed pathway in which one singlet is converted into two triplet excitons in organic chromophores.^{10,11} It is commonly phrased as



Here, ${}^1(TT)$ and ${}^1(T\cdots T)$ denote correlated triplet pair intermediates.¹² Initially, the ${}^1(TT)$ state is formed, which can be described as two electronically coupled T_1 states with an overall singlet spin. Upon spatial separation of this pair, electronic coherence is lost while maintaining spin correlation.^{13–17} Subsequent spin relaxation results in two free triplets. In general, SF occurs to a significant extent only if the S_1 state exhibits approximately twice the vertical transition energy of one T_1 state.¹⁰

A notable example is found in the case of tetracene, as temperature-dependent measurements suggest a coherent formation of the 1TT , while subsequent dissociation into two triplets is thermally activated.^{18–22} In contrast, favorable energetics in larger polyacenes result in exothermic SF,

which triggered studies on this class of molecules.^{11,14,15,23–40}

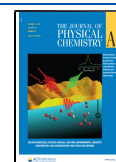
These showed that chemical modification has a significant impact on SF,^{36–38} which can be traced back to both morphological and electronic effects. First, altering the molecular structure usually results in changes in the crystal packing, which in turn affects intermolecular coupling essential for SF.³⁸ Second, heteroatoms have a direct impact on the energy levels of the molecule. For example, introducing nitrogen atoms into the aromatic core has shown to increase the diradical character which in turn lowers the lowest triplet state in energy.^{38,41–43} Beside polyacenes, perylene-based compounds have also been studied with respect to SF.^{44–52} While the energy levels obtained for single perylene-based molecules imply favorable energetics for SF, stabilization of the lowest singlet state, S_1 , in molecular crystals results in slow or even absent SF.^{49–52}

Based on this knowledge, we propose a new group of molecules as possible SF candidates, the tetraazaperopyrenes (TAPPs, Figure 1). According to the aforementioned design strategies, TAPPs are expected to have lower T_1 (vertical) transition energy, allowing for an efficient SF process. In

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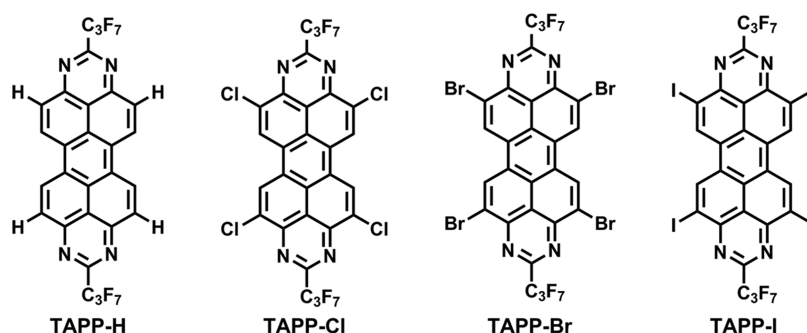


Figure 1. Molecular structures of the tetraazaperopyrenes (TAPP-X) investigated in the present work.

addition, the available synthetic routes allow for a plenitude of derivatives, leading to versatile electronic properties.^{53–60} In particular, halogenation has been shown to systematically tune the gap between the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO), the fluorescence quantum yield, and charge carrier mobilities.^{58–60} On the other hand, the substitution does not have a strong influence on the packing in the molecular crystal. This is especially notable for TAPP-Cl and TAPP-Br, which exhibit a face-to-face packing with short interplanar distances of 3.38 and 3.39 Å, respectively, and an almost equal horizontal and vertical displacement.⁵⁸ This mitigates morphological effects so that differences observed in dynamics mainly originate from electronic contributions.

2. METHODS

2.1. Sample Preparation. The TAPP derivatives were synthesized following procedures described in the literature.^{53,54,61} For studies in solution, the compounds were dissolved in acetonitrile (Sigma-Aldrich, ≥99.8%). Using the extinction coefficients reported previously,⁵⁸ the concentrations of TAPP-H, TAPP-Cl, TAPP-Br, and TAPP-I were determined to be 13.7, 9.5, 6.9, and 5.8 μmol L⁻¹, respectively. Stationary spectra were measured in a 2 mm quartz cuvette (Hellma) in a UV-2600 absorption spectrometer (Shimadzu). Thin-film samples were prepared on pre-cut glass substrates of dimensions 2.5 cm × 2 cm × 0.05 cm. They were cleaned by sonication in de-ionized water, acetone and 2-propanol for 10 min each, and blow-dried using a nitrogen gun. To enable a homogeneous surface, a 50 nm thick layer of cross-linked benzocyclobutene (BCB) (Dow CYCLOTENE 3022-35) was deposited via spin-coating. For this purpose, BCB precursor was diluted with mesitylene in a volumetric ratio of 1:6 after which it was spin-coated (500 rpm for 3 s followed by 8000 rpm for 60 s) and cross-linked for 60 s at 290 °C in a nitrogen glovebox. Subsequently, solutions were prepared by dissolving in *o*-xylene (Sigma-Aldrich, anhydrous, 97%) with a concentration of 5 mg mL⁻¹. The solutions were then sonicated for 5 min and heated at 110 °C for 15 min to ensure complete dissolution of the molecules. Thin films were prepared via a solution-based, meniscus-guided coating technique called zone casting.⁶² Solutions were maintained at 110 °C and cast on glass/BCB substrates heated to 130 °C to yield films with ribbon-like features (Figure S1). UV/vis spectra of thin films were measured using a Cary 6000i absorption spectrometer (Agilent). Thin films of TAPP-H and TAPP-I prepared by this technique did not achieve the suitable optical quality required for spectroscopic measurements.

2.2. Transient Absorption. Transient absorption measurements up to a probe delay of 8 ns were performed using a commercially available pump-probe setup (HELIOS, Ultrafast Systems). To generate the pump pulse, the fundamental beam at a center wavelength of 800 nm generated by a Ti:Sa amplified laser system (Astrella, Coherent, 4 kHz) was used as pump in an optical parametric amplifier (TOPAS Prime, Light Conversion). The samples were excited with 200 nJ pulse energy, unless stated otherwise, with an approximate spot size of 300 μm. White light supercontinuum generated in sapphire was used as probe. The relative polarization for pump and probe beam was set to 54.7° (“magic angle”) to account for molecular reorientation dynamics in solution and to 0° in thin-film measurements. The instrument response function was typically 90–100 fs. For longer pump-probe delays on the μs timescale, the pump pulse was generated as described above but probe pulse was implemented using an electronically delayed supercontinuum laser (EOS, Ultrafast Systems). Here, the instrument response function shows an FWHM of about 350 ps.

2.3. Fluorescence Measurements. Steady-state fluorescence measurements in solution were carried out in a JASCO FP 8200 spectrofluorometer using a 1 cm quartz cuvette. All fluorescence spectra were corrected for instrumental factors as described earlier.⁴⁰ Fluorescence lifetime measurements were carried out using a Lifespec II (Edinburgh Instruments) time-correlated single photon counting (TCSPC) spectrometer using an EPL-435 diode laser (Edinburgh Instruments) with a pulse width of 100 ps at 435 nm (2.85 eV) and selectable repetition rate.

Thin-film samples were excited at 450 nm (2.76 eV) by the spectrally filtered output of a pulsed supercontinuum laser source (Fianium WhiteLase SC400, 20 MHz repetition rate, ≈10 ps pulse width). A 20× objective was used to focus the excitation laser and collect the PL. The emitted light was dispersed by an Acton SpectraPro SP2358 spectrograph (grating 150 lines mm⁻¹) and detected by a liquid nitrogen-cooled InGaAs line camera (Princeton Instruments OMA V). Scattered laser light was blocked by a longpass filter (495 nm cut-on wavelength). For time-resolved PL measurements, the output of the spectrograph was focused onto a silicon avalanche photodiode (Micro Photon Devices). PL decay traces were recorded using PicoHarp 300 (PicoQuant) single photon counting electronics and fit using Symphotime 64 software.

2.4. Computational Details. All quantum chemical density-functional theory (DFT) and time-dependent DFT (TDDFT) calculations were performed with the Turbomole program package,⁶³ employing the PBE0 functional, grid 4 and

the def2-TZVPP basis.⁶⁴ Convergence criteria were tightened to 10^{-9} for both the energy ("scfconv") and the density ("denconv") as well as for the response vectors ("rpaconv"). All geometry optimizations, both in the electronic ground state and electronically excited states, were carried out using weight derivatives and in vacuum, i.e., without environment effects. Selected single-point calculations were carried out using the conductor-like screening model (COSMO), with dielectric constants of $\epsilon = 8.93$, 20.7, and 37.5 to simulate dichloromethane, acetone, and acetonitrile, respectively. Adiabatic transitions are obtained as the energy difference of the total state energy of the excited state at the optimized excited-state geometry and the total state energy of the ground state computed at the optimized ground-state geometry. If the zero-point vibrational energy (ZPVE) is contained, such an adiabatic energy transition is also denoted 0–0 transition to emphasize that the transition occurs among the vibrational ground states of each electronic state. An analysis of the excited states according to the Platt scheme is not achieved in the present work as it might be ambiguous for nitrogen-substituted hydrocarbons.^{65,66}

3. RESULTS

In the following, first, the absorption and emission spectra of TAPP molecules in solution and zone-cast thin films are presented. This enables the quantitative assignment of the ground-state bleach, which can be used to determine triplet yields in SF.^{44,51} Time-resolved spectroscopy measurements, by transient absorption and time-correlated single photon counting were performed to disentangle the dynamics of bright singlet states and dark, mostly triplet, states after photoexcitation. The experimental observations, supported by quantum chemical calculations, reveal the role of atom substitution on triplet generation and annihilation pathways in solution and thin film.

3.1. Stationary Spectra. The stationary absorption spectrum of TAPP-Cl has a maximum at 466 nm (2.66 eV) and a pronounced vibronic progression, with three well-resolved peaks separated by about 167 meV. The emission spectrum of the sample is a mirror image of the absorption spectrum with a pronounced peak at 476 nm (2.60 eV), corresponding to a Stokes shift of 56 meV (Figure 2a). The three other compounds exhibit similar features in absorption and emission spectra, i.e., vibronic progression and slight Stokes shift (Figures 2b and S2). The latter has been previously assigned to the strong contribution of the 0–0 transition.⁵⁹ Additionally, a systematic bathochromic shift is observed between the unsubstituted TAPP-H and its halogenated counterparts, i.e., Cl < Br < I, which is reproducible by quantum chemical calculations (Figure S2). The absorption and emission maxima do not exhibit significant solvatochromic effects (Figure S3). Therefore, we conclude that the electronic ground state and excited states of the sample are equally nonpolar, in agreement with previous reports on the hydrocarbon analogue of the TAPPs, peropyrene.⁵² To assess the effect of crystallization on TAPP-Cl and TAPP-Br energetics, the absorption and emission spectra of the corresponding zone-cast film are compared in Figure 2. The UV–vis absorption spectra of both zone-cast films exhibit a red shift of about 250 meV as well as a broadening of the peaks. Additionally, the relative intensity of the second peak in the emission spectra has increased in thin films compared to those

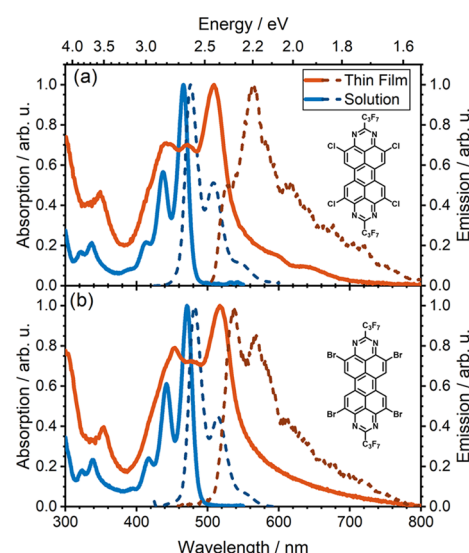


Figure 2. Stationary absorption (solid lines) and emission spectra (dashed lines) of (a) TAPP-Cl and (b) TAPP-Br. For both molecules, the spectra of zone-cast samples (orange) show a red shift of about 250 meV compared to measurements in acetonitrile (blue). The insets show the respective molecular structure.

in solution. For TAPP-Br, this ratio almost doubles from 0.45 to 0.87, while it increases by about a factor of 4 from 0.51 to 2 in TAPP-Cl, where the second peak at 565 nm (2.19 eV) becomes the dominant transition.

3.2. Time-Resolved Measurements. **3.2.1. Dynamics of Molecules in Dilute Solution.** In this section, we present the results from short-time (0–8 ns) and long-time (0–3 μ s) TA measurements in solution. The spectral evolution of all TAPP molecules dissolved in acetonitrile is very similar. The data are shown exemplarily for TAPP-Cl and TAPP-Br in Figure 3, which will be compared with the respective thin-film measurements in later sections. Initially, a pronounced

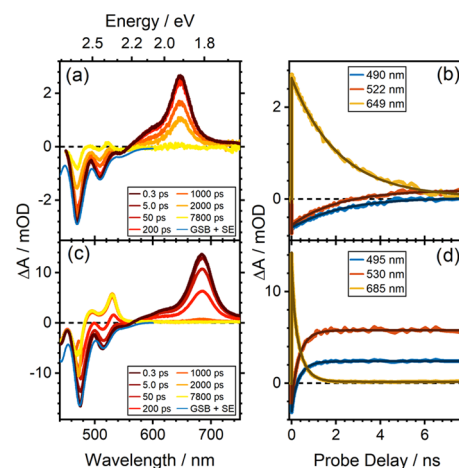


Figure 3. Transient absorption of (a, b) TAPP-Cl and (c, d) TAPP-Br in acetonitrile. Left: Transient absorption spectra at selected probe delays for (a) TAPP-Cl and (c) TAPP-Br with overlapped sum of GSB and SE (blue). The latter was obtained by adding the ground-state absorption and the stimulated emission cross section with manually weighted factors to match the negative signal. Right: selected kinetics with corresponding fit (black line) for (b) TAPP-Cl and (d) TAPP-Br, obtained by a global biexponential fit. The corresponding rate constants are collected in Table 1.

Table 1. Rate Constants of TAPP-X in Acetonitrile Obtained by a Global Fit of Transient Absorption and TCSPC Data at Maximum Intensity^a

molecule	short-time TA (0–8 ns)		long-time TA (0–3 μ s)		photoluminescence
	$k_1/10^9 \text{ s}^{-1}$	$k_2/10^6 \text{ s}^{-1}$	$k_2/10^6 \text{ s}^{-1}$	$k_3/10^6 \text{ s}^{-1}$	$k_{\text{PL}}/10^6 \text{ s}^{-1}$
TAPP-H	16.8 ± 0.1	491 ± 2	420 ± 11	6.29 ± 0.20	467 ± 1
TAPP-Cl	1.57 ± 0.07	478 ± 3	413 ± 2	3.56 ± 0.03	430 ± 1
TAPP-Br	12.3 ± 0.3	2667 ± 21	n.a.	3.73 ± 0.01	2762 ± 46
TAPP-I	424 ± 158	$(34.5 \pm 1.0) \times 10^3$	n.a.	3.64 ± 0.01	$(20.8 \pm 0.4) \times 10^3$

^aThe TA signal was fitted with a biexponential decay and an offset, while photoluminescence experiments were fitted with a monoexponential decay.

excited-state absorption (ESA) band on the red side of the spectrum is observed, with an additional intense negative signal assigned to ground-state bleach (GSB) and stimulated emission (SE). This signal can be reconstructed by adding the ground-state absorption spectrum and the stimulated emission cross section, the latter of which is obtained by multiplying the emission spectrum with λ^4 .⁶⁷ The weighting of both contributions was adjusted to optimally overlap with the TA negative signal. The ESA and SE bands vanish within few ns, while a new blue-shifted ESA appears around 525 nm (2.36 eV). This spectral feature exhibits two vibronic peaks and lives beyond the time range of this measurement. While this spectral evolution is similar for all derivatives, the spectral positions of the ESA bands are different. In line with stationary absorption spectra, the ESA bands also experience a bathochromic shift with increasing substituent size. The shift is more pronounced for the earliest ESA: The maximum of the transient absorption spectra of TAPP-Cl and TAPP-Br at 0.3 ps delay (Figure 3a,c) are separated by 100 meV from 649 to 685 nm (1.91–1.81 eV). Conversely, the maxima at 7.8 ns probe delay are located at 522 and 530 nm (2.38 and 2.34 eV), for TAPP-Cl and TAPP-Br, respectively, corresponding to a shift of 36 meV. Besides, the relative signal intensity of the blue-shifted maxima at late and red-shifted ESA at early probe delays is significantly larger for TAPP-Br $\approx$ 41% compared to those observed for TAPP-Cl $\approx$ 5.7%. This can be assigned to significantly higher fluorescence quantum yield in TAPP-Cl, as indicated by its high fluorescence quantum yield of 78%, compared to 12% in TAPP-Br observed in previous experiments.⁵⁸ This ratio is quantitatively consistent with the observed ESA intensities.

While the spectral evolution is similar for both compounds, the involved time scales differ significantly (Figure 3b,d). The formation of the second species takes place over the whole course of measurement (8 ns) for TAPP-Cl, whereas this process is already completed within 2 ns for the brominated compound. This is quantified by a global fit using a biexponential decay with an offset (Table 1). The first rate constant (k_1) explains small spectral shifts, and it is required to obtain the best fit (see Figure S4 for residuals of different fittings). The second rate constant (k_2) is related to substantial changes in the spectral shape and is, therefore, due to the formation of a spectrally separated electronic species. This rate constant does not change for TAPP-H and TAPP-Cl and increases systematically for heavier halogen substituents, which is further supported by time-resolved photoluminescence (k_{PL} , Figure S5). The offset component in the short-time measurements (<8 ns), decays, however, on a much longer timescale (k_3 in Table 1 and Figure S6a).

3.2.2. Dynamics in Thin Films. In the following, the results of time-resolved measurements on thin films of TAPP-Cl and TAPP-Br are compared. These two compounds were chosen

due to their practically identical crystal structures, which are characterized by short interplanar distances of 3.38 and 3.39 Å, respectively, as well as a horizontal ($\approx$ 3.4 Å) and vertical displacement ($\approx$ 1.3 Å) of the molecular axes.⁵⁸ Thus, packing effects in the solid are minimized, resulting in differences observed in their dynamics to originate mainly from electronic contributions.

3.2.2.1. Time-Resolved Photoluminescence. As evidenced by the stationary spectra shown in Figure 2 (and Figure S2), the emission spectra of solution and thin film differ more significantly for TAPP-Br than for TAPP-Cl. This is further investigated for thin films by time-resolved photoluminescence measurements carried out at the vibronic peaks of the respective emission spectra (Figure 4). For TAPP-Cl, the

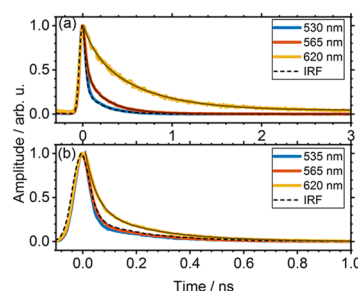


Figure 4. TCSPC traces of (a) TAPP-Cl and (b) TAPP-Br thin film at selected emission wavelengths. In both cases, the decay shows a red-shifted, long-lived emission, which can be fitted with a biexponential decay (Table S1). This second component is more pronounced at higher wavelengths and has a stronger contribution in TAPP-Cl compared to TAPP-Br.

photoluminescence (PL) decays within the time window of the instrument response function (IRF) for the emission at the blue edge. In turn, longer wavelengths show a systematic decrease of the decay rate (Table S1). For TAPP-Br, this effect is less pronounced and only notable at 620 nm (2.00 eV). The observations suggest that a second emissive species is present with a long-lived, red-shifted emission, which has a higher contribution in the overall emission in TAPP-Cl (cf. SI Section 4.2).

3.2.2.2. Transient Absorption. Like the TA data in solution (vide supra), the dynamics observed for TAPP-Cl and TAPP-Br thin films are similar (Figure 5). At 200 fs probe delay, the transient absorption spectra show a broad ESA band in the red part of the spectral window >600 nm (<2.07 eV) and an additional narrower ESA overlapping with the GSB at 490 nm (2.53 eV) for TAPP-Cl (Figure 5a,c). These features are slightly red-shifted by about 60 meV for TAPP-Br. Furthermore, the relative amplitude of the narrow ESA peak compared to the red-shifted ESA is significantly higher in

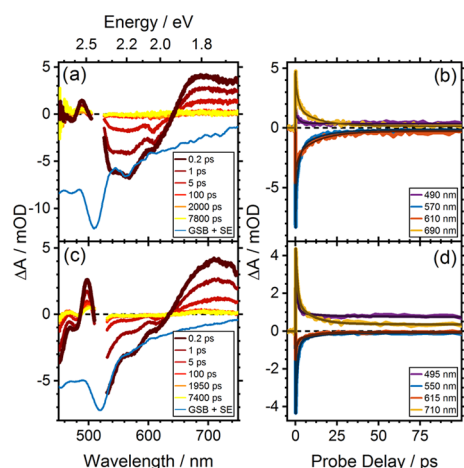


Figure 5. Transient absorption of (a, b) TAPP-Cl and (c, d) TAPP-Br thin films. Transient absorption spectra at selected probe delays for (a) TAPP-Cl and (c) TAPP-Br. Selected kinetics with corresponding fit (black line) for (b) TAPP-Cl and (d) TAPP-Br, obtained by a global biexponential and triexponential fit, respectively. In the analysis of the TA data of TAPP-Cl and TAPP-Br thin films, the spectral region ± 10 nm around the excitation wavelength of 515 and 520 nm (2.41 and 2.38 eV), respectively, was removed due to excessive scattering.

TAPP-Br (62%) compared to TAPP-Cl (36%). An additional negative signal is observed above the excitation wavelength, which can be reconstructed by a combination of GSB and SE (as performed for TA data in solution, *vide supra*). Within the first 5 ps after photoexcitation, a signal decay is observed over the whole spectral range. This decrease is homogeneous for both ESA bands. Contrarily, the SE contribution shows a slower decay for wavelengths around 610 and 615 nm (2.03 and 2.02 eV) for TAPP-Cl and TAPP-Br, respectively (Figure 5b,d). Subsequently, the red-shifted ESA band vanishes almost completely, whereas the narrow peak retains about one-third of its initial signal intensity. For TAPP-Cl, no further spectral evolution is observed on this timescale, whereas the narrow ESA peak exhibits a slight red shift in TAPP-Br (Figure S7 and Section 6, SI).

This difference in TAPP-Cl and TAPP-Br is reflected in the number of exponential decays needed to globally fit the dynamics (Table 2): The ps–ns dynamics for TAPP-Cl can be

Table 2. Rate Constants Obtained by a Global Fit of the Transient Absorption Data in Thin Film^a

molecule	$k_1/10^9 \text{ s}^{-1}$	$k_2/10^9 \text{ s}^{-1}$	$k_3/10^9 \text{ s}^{-1}$
TAPP-Cl	565 ± 3	74.6 ± 0.4	n.a.
TAPP-Br	893 ± 8	105 ± 1	1.88 ± 0.06

^aThe TA fit indicates a triexponential decay and an offset.

fitted with two independent exponential decays and an offset, whereas an additional exponential component (three in total) is needed for TAPP-Br (Figure S8). Due to the high signal intensity of the long-lived component in TAPP-Br, it can be tracked on the microsecond timescale, where it exhibits a homogeneous decay (Figure S6b).

3.2.2.3. Vibrational Coherence. The TA measurements performed on TAPP-Cl and TAPP-Br show the presence of oscillatory features at early pump-probe delays. These periodic features correspond to the oscillations of a vibrational

wavepacket generated by the pump pulse. These oscillations are more prominent in the TA signal of TAPP-Br compared to TAPP-Cl (Figure S9), which is attributed to the more crystalline morphology of the TAPP-Br film (Figure S1) and the associated better signal-to-noise ratio (less scattering). The oscillatory contributions in the thin-film signal of TAPP-Br can be extracted by performing a fast Fourier transform (FFT) on the fit residuals, starting at a probe delay of 200 fs to avoid artifacts from the coherent spike. The resulting spectrally resolved FFT spectrum shows five main contributions at about 20, 50, 125, 150, and 205 cm^{-1} (Figure 6a), with their respective amplitudes being strongest close to the excitation wavelength. These frequencies show no shift across the probe spectrum, as evidenced by cuts at selected probe wavelengths (Figure 6b).

To gain further insight into the origin and possible role of the observed vibrations in the excited-state relaxation, the dependence of their amplitude on probe delay is evaluated. A common approach would be a sliding-window FFT.²² This is, however, not feasible here due to the long vibrational period of the 20 cm^{-1} mode in relation to its dephasing rate. Thus, the kinetic trace at the maximum of the narrow ESA peak (495 nm, 2.50 eV) is fitted with the sum of five sinusoidal functions, a monoexponential decay and an offset (Figure 6c, see also SI Section 10). By subtracting the exponential decay, the pure oscillatory component is obtained (Figure 6d). The residual of this fit shows no periodic structure, which proves that no further oscillatory component is present within the experimental resolution (Figure 6e). The fit further allows for the extraction of individual contributions of the oscillatory component, depicted in Figure 6f. Their respective frequencies of 19.9, 52.5, 126, 143, and 206 cm^{-1} are in good agreement with the FFT spectra. Additionally, dephasing rates of the first four vibrations show an expected systematic increase with increasing frequency, going from $(36.6 \pm 2.0) \times 10^{10} \text{ s}^{-1}$ to $(76.9 \pm 23.7) \times 10^{10} \text{ s}^{-1}$.⁶⁸ In contrast to this trend, the remaining vibration of 206 cm^{-1} dephases with $(49.8 \pm 29.7) \times 10^{10} \text{ s}^{-1}$. However, due to its low amplitude, this rate constant is determined with a large error.

3.2.3. Computational Results. Vertical transition energies are collected in Tables 3 and 4 for the R-TAPP-Br compound with $R = \text{C}_3\text{F}_7$ and $R = \text{H}$, computed using full time-dependent DFT (TDDFT) and the Tamm–Dancoff approximation (TDA). The tables reveal a deviation of these two quantum chemical approaches of about 0.2–0.3 eV and, as well understood, that the substituent R has only a small effect upon the vertical transition energies. The vertical emission energy is reduced by about 0.15 eV compared to the vertical absorption energy.

As shown previously,⁵⁹ the TAPP compounds exhibit a strong allowed 0–0 transition due to the similar geometries of ground and excited states so that the pure vertical transitions yield large deviations when compared directly to experimental spectra. Hence, for a quantitative analysis, computed adiabatic or 0–0 transitions (Table 4) should rather be compared to the experimental data collected in Table 5. The 0–0 transition is computed to be about 2.5 eV using TDDFT and the harmonic approximation, which is in sufficient agreement with the measured value of about 2.6 eV in solution and 2.3 eV in film in view of the different environments. In Table 4, the values without parentheses refer to adiabatic transitions without zero-point vibrational energy (ZPVE) correction, while the values in parentheses refer to 0–0 transitions including the ZPVE in the

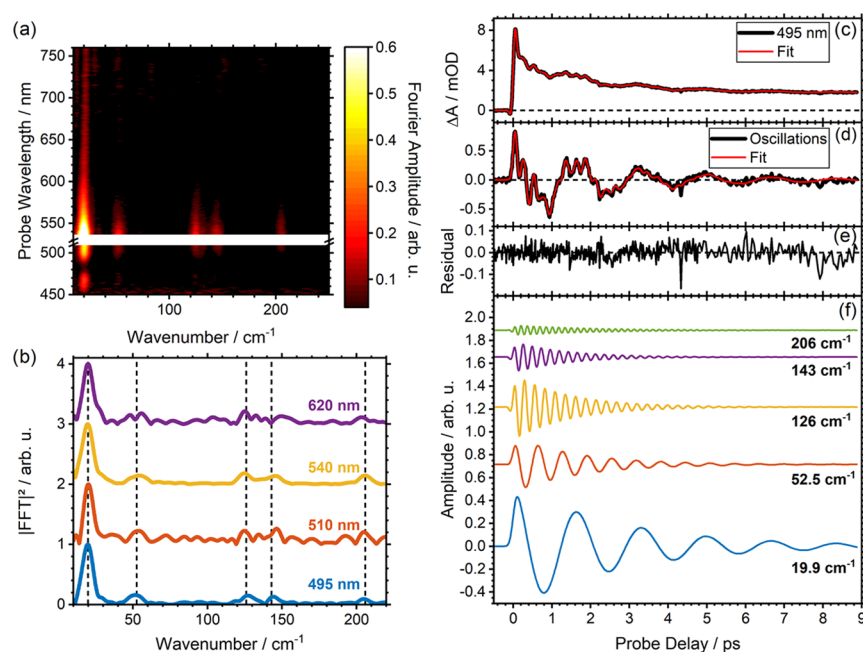


Figure 6. Vibrational coherence observed in the TAPP-Br thin film. (a) Spectrally resolved FFT spectrum of the residuals obtained by a global fit of the initial 10 ps. (b) FFT spectra at selected wavelengths. (c) Fit of the kinetics at 495 nm (2.50 eV) with exponential decay and five oscillatory components. (d) Oscillatory component obtained by subtracting contributions of the coherent spike and population dynamics. The fit accurately reproduces the experimental data, as evidenced by the structureless residuals in (e). The frequencies of the individual components (f) agree with the FFT spectra, as indicated by the dashed lines in (b). Results are collected in Tables S1–S3.

Table 3. Vertical Transition Energies in eV for C₃F₇-TAPP-X Using a Polarizable Continuum Model (COSMO) with a Dielectric Constant of 37.5

state	TDDFT geometry	TAPP-H	TAPP-Cl	TAPP-Br	TAPP-I
S ₁	S ₀	2.89	2.62	2.58	2.46
	S ₁	2.66	2.40	2.36	2.27
T ₁	S ₁	1.24	1.08	1.10	1.10
T ₂	S ₁	2.71	2.45	2.40	2.26

Table 4. Vertical and 0–0 Transition Energies in eV for H-TAPP-Br Using the Tamm–Dancoff Approximation (TDA) and Full TDDFT^a

state	transition	TDA	TDDFT
S ₁	absorption	2.87	2.68
	emission	2.70	2.41
	adiabatic (0–0)	2.81 (2.73)	2.57 (2.50)
T ₁	absorption	1.68	1.40

^aValues without parentheses are without zero-point vibrational energy (ZPVE) correction; values in parentheses include the ZPVE. For these compounds, only the adiabatic and 0–0 transitions should be compared directly to experimental data.

Table 5. Experimental Absorption and Emission Maxima in eV for C₃F₇-TAPP-X in Acetonitrile (ACN) and Thin Film

	TAPP-H	TAPP-Cl	TAPP-Br	TAPP-I
absorption (ACN)	2.85	2.66	2.63	2.54
emission (ACN)	2.81	2.61	2.57	2.46
absorption (film)		2.44	2.40	
emission (film)		2.36	2.32	

harmonic approximation. It shows that the ZPVE reduces the adiabatic transition energies slightly by about 0.07 eV.

Furthermore, the data correctly predict the minor impact of the solvent polarity on energy levels (see SI Section 13). Additionally, the computed vibrational frequencies of the ground state as well as first excited singlet and triplet states show only minor deviations below 250 cm⁻¹ (Table S9).

4. DISCUSSION

4.1. Molecules in Dilute Solution. In solution, a sequential kinetic model using rate constants obtained by the global fit (vide supra) was applied to obtain the species-associated difference spectra (SADS) of the species involved. All four compounds show similar features for the first SADS, i.e., negative contributions from GSB and SE as well as a red-shifted broad ESA band (Figure 7a). Furthermore, the relative amplitude of the ESA in comparison to GSB/SE signal decreases with substituent size. The second SADS show the same spectral features for all molecules, albeit with the amplitude lowered by about one-third (Figure 7b). This spectral similarity suggests that the first two SADS originate from the same electronic state. Consequently, they are assigned to an initially excited S₁' state, which undergoes structural relaxation to the S₁ state with a rate characterized by *k*₁. This change in molecular geometry results in a redistribution of oscillator strengths for the probed S₁ → S_n transitions, which explains the change in amplitude.

Comparing the red-shifted ESA band in the SADS for the individual molecules shows a gradual shift of about 100 meV going from TAPP-H to TAPP-I. This systematic trend is observed for the stationary spectra as well (vide supra), for which it stems from the expansion of the π -system by the halide substituents.⁵⁸ Consequently, it is concluded that the red shift of the S₁ → S_n transition has the same origin. In contrast, the final SADS (Figure 7c) shows distinctively different spectral features. In agreement with the photo-

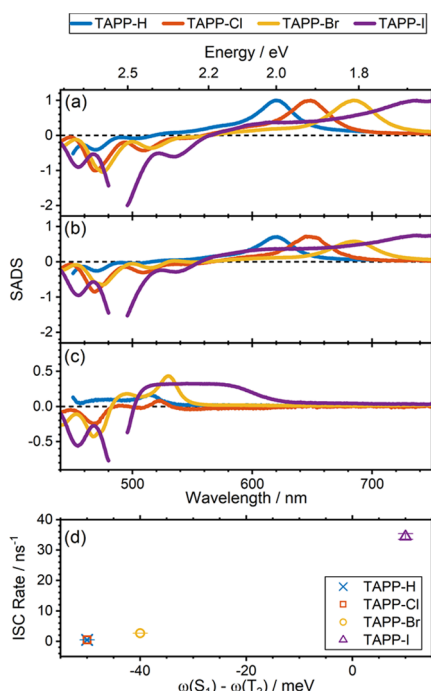


Figure 7. Comparison of the SADS of the four TAPP molecules in solution normalized for the maximum of the S_1 SADS. (a) The initial S_1' SADS show a red shift with substituent weight, as observed for stationary spectra. (b) The intermediate SADS have the same spectral shape as the S_1' SADS with their respective intensity decreased by one-third and are thus assigned to the geometrically relaxed S_1 state. (c) The final SADS significantly differ from the first two spectra. Due to their long lifetime on the microsecond timescale, they are assigned to the T_1 state formed via ISC. (d) The measured rate constant for ISC in solution shows a direct correlation with the calculated energy detuning between the states S_1 and T_2 (Table 3), indicating that the ISC occurs between these two states, followed by a fast internal conversion from the T_2 to the T_1 state.

luminescence decay, no SE is present and a blue-shifted ESA band is observed, partially overlapping with the GSB (Figure 7c).

In general, the SADS show two vibronic peaks, apart from TAPP-I, for which a broad signal between 525 and 570 nm (2.36 and 2.18 eV) is present. For the other three compounds, the main peaks show a gradual red shift from 515 nm (2.41 eV, TAPP-H) to 530 nm (2.34 eV, TAPP-Br). As this last species is formed with the rate constant present in the photoluminescence measurements (Table 1), the spectrum has to correspond to an optically dark species, most likely the T_1 state. This assignment is further justified by the mono-exponential decay with a rate constant $<7 \times 10^6 \text{ s}^{-1}$.

The formation of a triplet state in solution can in principle occur via ISC or, due to the high S_1 – T_1 energy gap (Table 3), energy transfer to $^3\text{O}_2$.⁶⁹ For the latter, an upper limit can be calculated using the quenching rate of oxygen⁶⁷ and its concentration in acetonitrile.⁷⁰ As the resulting estimation $80.7 \times 10^6 \text{ s}^{-1}$ is significantly lower than the observed values in all four molecules, this reaction channel is ruled out. The further possibility of a direct, diffusion-controlled SF process can be excluded due to the low concentration of the samples ($\ll 0.1 \text{ mmol L}^{-1}$).^{27,69,71} Thus, it is concluded that the triplet state is formed via an ISC process with the rate constant k_2 . This is supported by the systematically increasing rates with substituent size for the three halogenated TAPPs and the

concomitant decrease of the fluorescence quantum yield.⁵⁸ An exception to this trend is the unsubstituted TAPP-H, which exhibits a slightly faster triplet formation compared to TAPP-Cl. However, this effect can be rationalized by the energy detuning of the involved electronic states: The initial vibrationally relaxed S_1 and the T_2 state differ only in a few tens of meV for all compounds (Table 3). Within the accuracy of the method, TAPP-H and TAPP-Cl have the same value of -50 meV , in agreement with the similar rate constants for both compounds (Figure 7d). Consequently, the triplet formation is assigned to ISC from the S_1 to the T_2 state, which is followed by a rapid internal conversion (faster than the ISC rate) to the observed T_1 state.

4.2. Dynamics in Thin Films. The transient spectra of TAPP-Cl and TAPP-Br thin films at early probe delays show ESA features in the same spectra regions, where the $S_1 \rightarrow S_n$ and $T_1 \rightarrow T_n$ transitions have been identified in solution. For a more detailed comparison, steady-state absorption spectra and stimulated emission cross section were combined to subtract the negative signal features of the respective T_1 SADS (Figure 8a,b).

Two major features can be observed. First, the singlet transition in the thin film at 200 fs probe delay is red-shifted and broadened compared to solution (see region 600–750 nm (2.07–1.65 eV) in Figure 8a,b). Second, the blue-shifted ESA of the thin-film TA closely matches the T_1 SADS (see region below 550 nm (2.25 eV) in Figure 8a,b). This is especially notable for TAPP-Br (Figure 8b), for which the spectrum is identical except for a blue shift of 150 meV. Consequently, we assign the red-shifted ESA features to a transition in the singlet manifold and the blue-shifted ESA features to a transition of a triplet species, both of which are formed in thin films on a timescale faster than the experimental resolution (90–100 fs).

This raises the question of the nature of the initially present triplet signal. As the formation of a triplet state via ISC is not expected to be accelerated by several orders of magnitude in the film compared to solution, it is assigned to the correlated triplet pair state¹(TT). Such a fast formation agrees with previous results for coherent ¹(TT) formation observed, e.g., in hexacene,³¹ rubrene,^{16,35} TIPS-Pn derivatives,^{14,15} and perylene diimide.⁴⁴ To further support this assignment, photoluminescence measurements were carried out in the NIR spectral region. However, no emission was observed, which is attributed to a low intensity of the phosphorescence.

The feasibility of a coherent ¹(TT) formation is further supported by the computed state energies of the involved lowest singlet, S_1 , and lowest triplet, T_1 , states (Table 3). For isolated TAPP-Cl chromophores and in a zeroth-order model, the formal energy balance for the fission of the lowest singlet state S_1 (formally available energy of 2.40 eV) into two T_1 states (formally required energy of $2 \times 1.08 \text{ eV}$) at the relaxed excited-state S_1 geometry is positive by approximately 240 meV. Analogously, a calculation for TAPP-Br yields an excess energy of 160 meV. Due to energy conservation, these formal values are understood to be directly converted to vibrational energy. While indicating the possibility of SF, however, such a zeroth-order model neglects for example stabilization effects in thin film, which are typically stronger for the S_1 state than for the T_1 state.⁷² This will strongly influence SF and can change the energy balance from exothermic to endothermic, e.g., in the case of peropyrene and its 2,9-disubstituted derivatives.^{47,52} To quantify such stabilization effects, the red shift between the stationary absorption spectra in solution and thin film is

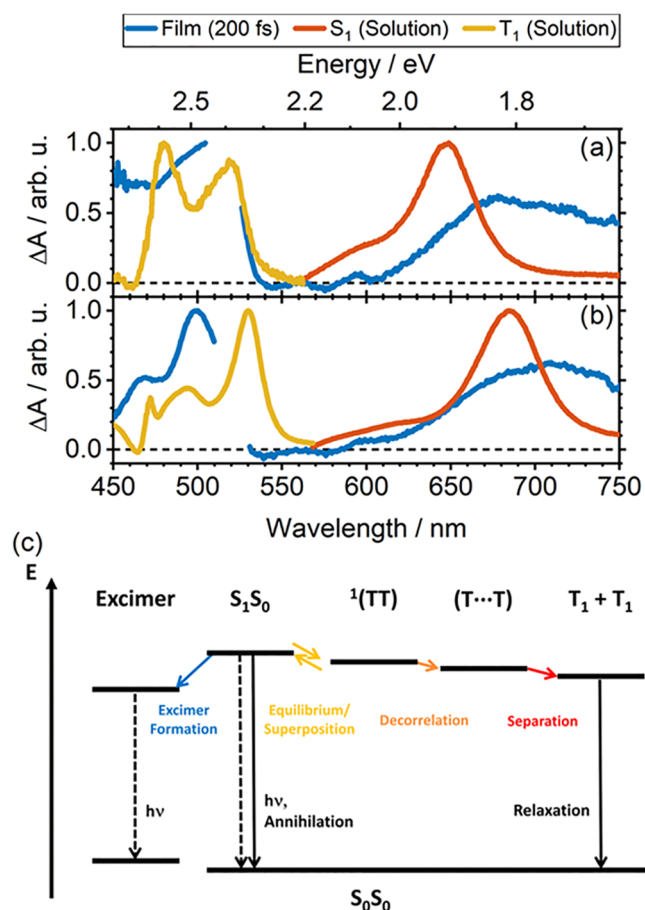


Figure 8. S_1 (orange) and T_1 (yellow) SADS of solution TA and transient spectra at 200 fs probe delay (blue) of thin films of (a) TAPP-Cl and (b) TAPP-Br. Negative contributions of GSB and SE were subtracted to allow for a comparison of the three spectra. The spectral features observed for thin film can be explained as a combination of singlet and triplet features observed in solution, indicating the presence of a triplet species being formed coherently with photoexcitation. (c) Jablonski-type diagram of the proposed kinetic model in thin films. The dashed arrows represent radiative transitions. Upon photoexcitation, the excited singlet state, S_1S_0 , is in an equilibrium with the $^1(TT)$ state (yellow). From here, two reaction channels are possible: (i) A long-lived excimer state is formed (blue), which experiences a slight stabilization compared to the initial excitation, resulting in a red-shifted emission (ii) The $^1(TT)$ state undergoes electronic decorrelation (orange) to yield the $(T\cdots T)$ state, subsequently followed by separation and relaxation. The absence of delayed fluorescence indicates that no recombination of the $(T\cdots T)$ state takes place.

assumed to monitor the stabilization of the S_1 state. As this estimation neglects a possible stabilization of triplets, it depicts a “worst case” scenario. Following this approach, the energy balance shifts to 15 and -74 meV for TAPP-Cl and TAPP-Br, respectively. This shows the energetic feasibility of SF in TAPP-Cl and TAPP-Br, especially when considering the observation of SF in endothermic systems, most prominently tetracene.^{22,73}

4.2.1. Interpretation of the Temporal Evolution. The interpretation of the temporal evolution of the transient spectra in thin films gives further insight into the evolution of the singlet and triplet states. Figure 8c summarizes in a Jablonski-type diagram the relaxation pathways discussed in the following.

Initially, both triplet and singlet features show a sharp drop in intensity within the first 10 ps, characterized by k_1 . This is attributed to second-order annihilation processes such as singlet–singlet annihilation, which is supported by the acceleration of the decay with increasing pump intensities (Figure S10). Another possible explanation for this behavior would be the independent decay of two heterogeneous populations originating from defects in the film. To match the observations of the homogeneous decay, however, this would require the unlikely assumption that these two heterogeneous populations decay with the same rate constant. A more plausible explanation for the simultaneous presence of singlet and triplet features is either a rapid interconversion or superposition of the S_1S_0 and the $^1(TT)$ state.⁴⁰ Evidence for this can be found in the time-resolved PL measurements (Figure 4). In both compounds, the photoluminescence decays within the experimental resolution independent of the emission wavelength (<100 ps) at the blue edge of the spectrum, indicating an efficient fluorescence quenching. Furthermore, a long-lived, red-shifted emission is observed. In analogy to the hydrocarbon parent compound of the TAPPs, peropyrene,⁵² this spectral feature is assigned to an excimer state.

Following the initial homogeneous spectral evolution, the second rate constant of k_2 corresponds to the singlet ESA decaying almost completely while the triplet feature retains its intensity (Figure 5; see also Figure S8 for the SADS). The change of the relative amplitude of singlet and triplet features means that both states are no longer interlinked. As no spectral change is observed for the triplet signature, it is concluded that the initially identified $^1(TT)$ state efficiently decorrelates, thus preventing a fast interconversion with the singlet. Nonetheless, the singlet ESA retains a small amplitude, which could be explained by a repopulation of the S_1 state. However, this would result in a delayed fluorescence, which is not observed. Thus, the remaining weak signal for the red-shifted ESA is assigned to a long-lived excimer (Figure 8c). This is supported by the inhomogeneous decay of the SE (Figures 5 and S8): Here, it is clearly notable that the signal in the spectral region around 610 and 615 nm (2.03 and 2.02 eV) for TAPP-Cl and TAPP-Br, respectively, decays slower than the remaining SE feature. These findings agree with the red-shifted, long-lived emission observed in PL measurements. The formation of the excimer could in principle occur directly upon photoexcitation or as a product of triplet fusion, as seen in perylene.^{33,45,74,75} However, as the intensity of the excimer emission both in PL and TA measurements directly correlates with the intensity of the triplet signal, it is concluded that the formation of the latter acts as a competing pathway to SF. This is in agreement with studies suggesting that excimer formation results in an energetic stabilization and thus unfavorable energetics for SF.^{33,45,74} This argument is consistent with the rate constant k_2 (Table 2), which encompasses both excimer formation and decorrelation: Assuming a similar excimer formation rate, the higher value for TAPP-Br corresponds to a larger population fraction undergoing decorrelation rather than excimer formation.

The product of electronic decorrelation could either be the spin-entangled $(T\cdots T)$ state or the direct formation of a pair of free triplet states ($2 T_1$). In the latter case, the free triplets could undergo relaxation to the ground state either via triplet–triplet annihilation or ISC. This case seems unlikely given the complex subsequent dynamics of the triplet signal in TAPP-Br,

for which one rate constant (k_3) is observed on the nanosecond (0–8 ns) and two additional rate constants on the microsecond timescale (Figure S6). Thus, the product of electronic decorrelation is assigned to the ($T\cdots T$) state, which undergoes spin relaxation with $k_3 = (1.88 \pm 0.06) \times 10^9 \text{ s}^{-1}$.

The overall efficiency of SF can be obtained by determining the yield of the free triplet state. In general, three approaches are conceivable. First, the concentration of the initial photoexcited state can be directly compared to the concentration of the triplet state. The former can be determined via the extinction coefficient known from stationary UV/vis spectra and the GSB intensity, while the extinction coefficient of the triplet state is typically approximated by triplet sensitization experiments. However, in the present case, the intensity of the GSB cannot be accurately determined due to the strong pump scattering within the experiment. Second, the triplet yield can be determined as the fraction of the SF rate k_{SF} and the total singlet decay rate k_{tot} . Besides k_{SF} itself, k_{tot} includes the rates of all radiative and non-radiative relaxations, e.g., fluorescence (k_{F}), internal conversion (k_{IC}), ISC (k_{ISC}), and competing reactions such as excimer formation (k_{Exc}). Thus, if all of those rate constants were known, the triplet yield would be available from

$$\Phi_{\text{T}_1} = 2 \frac{k_{\text{SF}}}{k_{\text{SF}} + k_{\text{F}} + k_{\text{IC}} + k_{\text{ISC}} + k_{\text{exc}}} \quad (2)$$

In TA experiments, the obtained experimental rate is equal to k_{tot} and as such does not a priori disclose details about the individual rates it is composed of. To extract these individual rates, further information is required which is not available. Consequently, an accurate estimation of the triplet yield is not feasible. As a third approach, the efficiency of SF can be determined by a comparison of the amplitudes of the SADS obtained for triplet pair and free triplet.^{38,40} As expected, the SADS for the triplet pair and free triplet show similar spectral features (Figure S8). However, they overlap the GSB region, resulting in contributions of the excimer-state dynamics to their respective amplitudes. As such, an exact ratio between SF and excimer formation is needed to accurately model the influence of the excimer. As this information is not available, this approach would introduce an unknown error to the triplet yield. Therefore, none of the three methods can be applied to determine the triplet yield quantitatively in the present work.

4.2.2. Intermolecular Vibrational Modes. A comparison of the FFT obtained from the thin film and the calculated ground-state vibrational frequencies for the monomer is made (Figure 9). In the latter, only bands with significant Raman intensities are displayed, leading to main contributions from modes located at 110, 140, and 205 cm^{-1} , which are also present in the experimental spectrum, and, therefore, are assigned as intramolecular modes. The assignment of the 205 cm^{-1} mode is further supported by IVS experiments in acetonitrile (SI Section 11). Deviations between calculations and experiment can be found for the modes at 40 and 90 cm^{-1} : While the former is 10 cm^{-1} below the 52 cm^{-1} mode, no experimental intensity is observed around 90 cm^{-1} . Additionally, no intensities are obtained for the intense experimental modes at 20 and 126 cm^{-1} , which indicates that these must arise due to molecular interactions only present in the respective dimer. Similar nonlocalized modes have been shown to be crucially involved in the coherent formation of

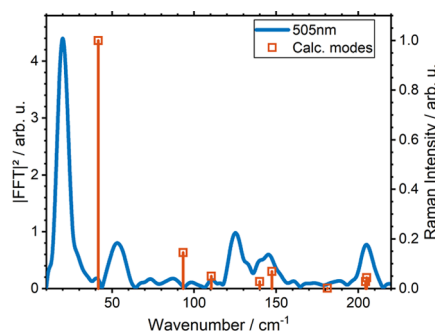


Figure 9. Comparison of the power FFT spectrum of TAPP-Br at a probe wavelength of 505 nm (2.46 eV) and calculated Raman intensities. For the latter, only modes with a nonzero intensity are shown in the selected range up to 220 cm^{-1} .

the $^1(\text{TT})$ state in other compounds.^{12,22,29,31,34,35,76,77} In the case of tetracene, this results in high SF efficiencies despite unfavorable energetics.²² For rubrene, vibrations break the symmetry within the molecular crystal, enabling an SF process that would be inactive otherwise.³⁵

Assignment of the three intramolecular modes (110, 140, and 205 cm^{-1}) to a single electronic state in TAPP-Br is, however, not straightforward. Usually, the spectral dependence, dephasing rates, frequency, and/or the phase of the oscillatory signal are considered to assign the respective Raman mode to a ground or an excited potential surface.⁷⁸ The modes observed for TAPP-Br show enhanced activity in spectral regions where the ground-state absorption, singlet and triplet manifold show a maximum (Figure 6), which makes their assignment in that regard not possible. Additionally, the computational modeling does not support significant frequency shift for each mode in the S_0 and S_1 and T_1 states (Table S9). This is observed in the experimental data as well (Figure 6). The different phase of each fitted oscillatory signal (Table S4) suggests, however, that they are originated in different states. The experimental dephasing rates obtained for the observed modes are within 30 to 80 $\times 10^{10} \text{ s}^{-1}$ (Table S4) and, thus, compatible with the decay rate of the $^1(\text{TT})$, which decays with $8.47 \times 10^{10} \text{ s}^{-1}$, as well as to a mode in the ground state. Based on these arguments, it is very likely that the experimentally observed vibrational modes originate from more than one potential surface by an impulsive Raman process generated by the pump pulse. A disentanglement of each contribution based only on the oscillatory features (vide supra) and thus a potential assignment to states involved in the SF process is ambiguous. This can be addressed in the future by applying higher-order time-resolved vibrational techniques sensitive to the excited-state manifold.⁷⁹

5. CONCLUSIONS

In the present work, the effect of halogenation on the dynamics of TAPP derivatives is investigated in solution and thin film. A combined experimental and computational ansatz shows that fundamentally different triplet formation mechanisms are observed in solution and thin films for four TAPP derivatives. In solution, two distinct electronic species with spectrally well-separated ESA bands are formed. Based on photoluminescence decay and the observed decay rates, these bands are assigned to the S_1 and T_1 states, respectively. For the formation of the triplet state, rapid ISC in all four TAPPs is observed, which indicates a direct correlation with the S_1 – T_2 energy detuning,

thus being a crucial factor for the rate constant. In thin films of TAPP-Cl and TAPP-Br, a comparison of transient spectra at early probe delays with SADS obtained from solution measurements allow for the separation of singlet and triplet signatures. Both features appear on the sub-100 fs timescale, suggesting an ultrafast formation of the $^1(\text{TT})$ state, with the latter dephasing on the picosecond timescale. The absence of delayed fluorescence shows that no recombination of the triplet pair takes place. Furthermore, the presence of a red-shifted emission with a lower decay rate compared to blue-shifted fluorescence indicates the competing formation of an excimer species, which is concomitantly identified in TA experiments. To summarize, the present work shows that azasubstitutions within the aromatic core lead to both qualitative and quantitative changes for SF and could thus be used as a powerful tool to fine-tune molecular properties in regard to SF dynamics.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpca.0c04852>.

Microscopy images, stationary spectra, residuals of the global multiexponential fit of TAPP-Cl and TAPP-Br in acetonitrile, time-resolved emission, long-time transient absorption, derivative spectroscopy, species-associated difference spectra, periodic oscillations, excitation intensity dependence, fit of the vibrational coherence, vibrational coherence in solution, calculated energy levels, and calculated vibration frequencies (PDF)

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Notes

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